Problems

Find the total capacitance when two capacitors of values 1000μF and 470μF connected in parallel.

Total capacitance $C = C_1 + C_2$

$C = 1000 + 470$

$C = 1470\mu F$.

1.11.11 Action of a Capacitor in DC Circuit

Figure 1.69 shows a capacitor connected in a DC circuit.

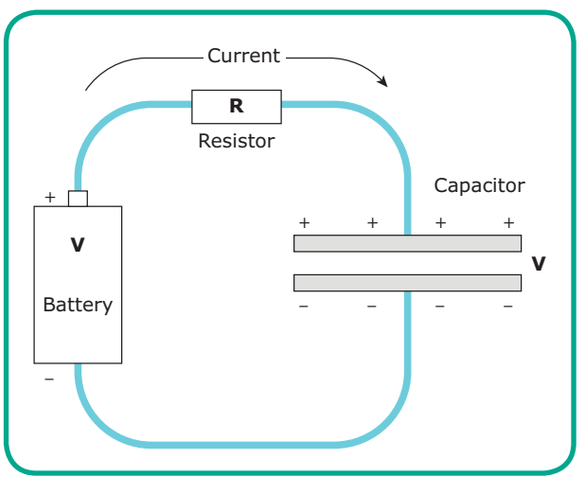


Figure 1.69 Action of a capacitor in DC circuit

When a capacitor is connected in DC circuit, the current flow will be high until it is charged fully.

When it is charged completely, the flow of current will be stopped and the voltage in the capacitor will be equal to the EMF in the battery. When a capacitor is connected in series with DC, it blocks the supply.

1.11.12 Action of a Capacitor in AC Circuit

Figure 1.70 shows a capacitor connected in an AC circuit.

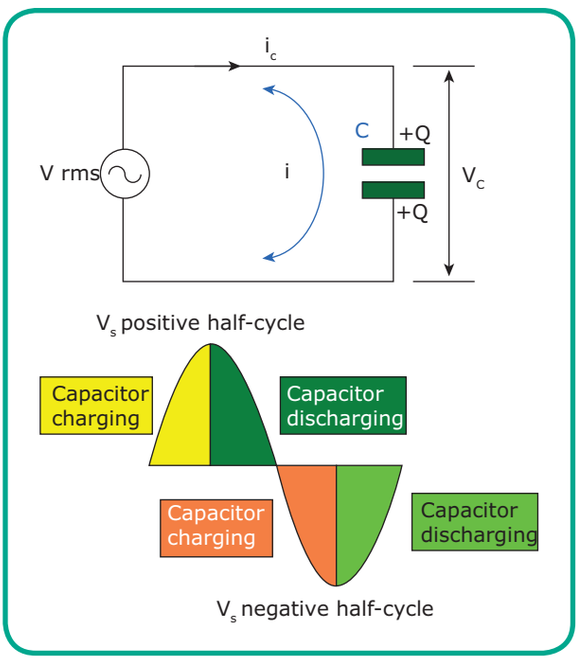


Figure 1.70 Action of capacitor in AC circuit

When a capacitor is used in AC circuit, it will be charged and gets discharged for every quarter cycle, since polarity is changed for every quarter cycle (i.e., 90°).

LEARNING OUTCOMES

After studying this Chapter, a student can understand the following

1. **Basics of electricity**
2. **Electrical parameters**
3. **Ohm's law**
4. **Kirchhoff's law**
5. **Working of Resistors**
6. **Working of Capacitors**

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